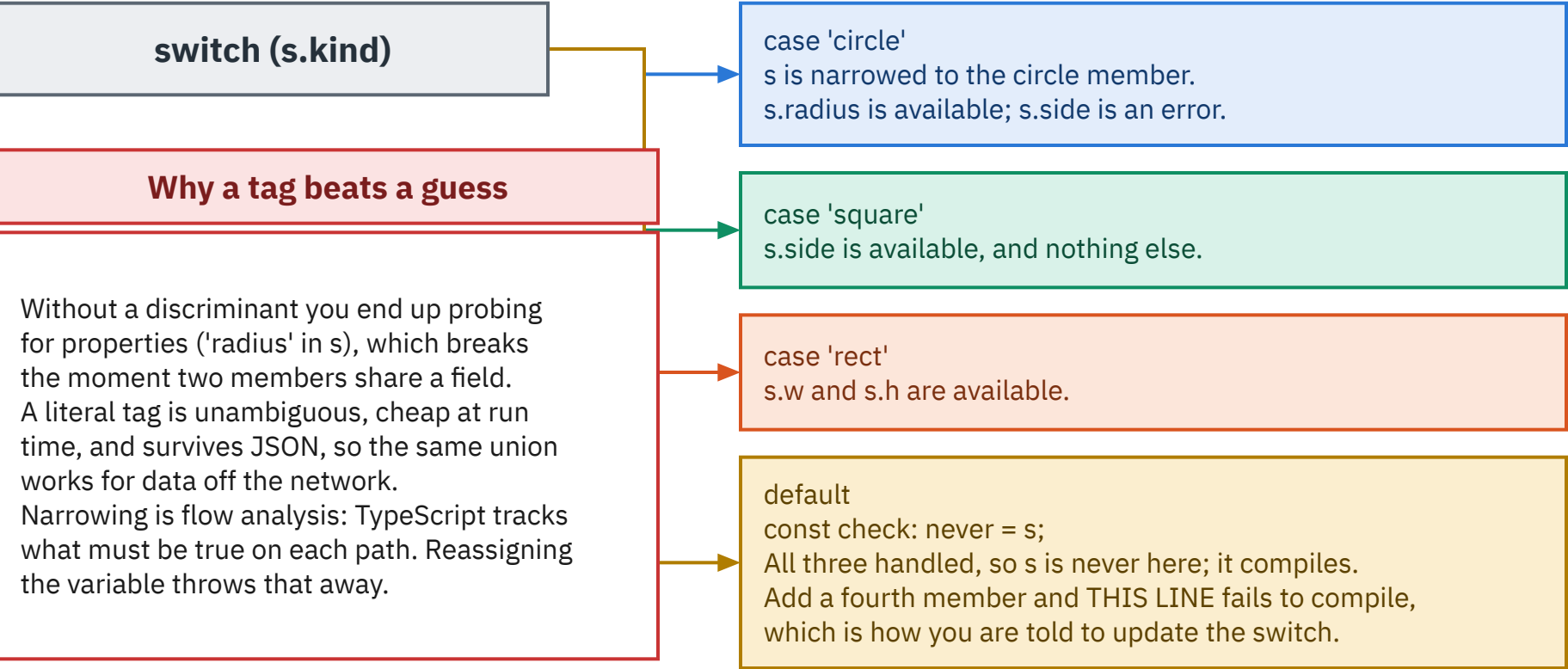


Narrowing and Discriminated Unions

Give every member a shared tag, switch on it, and TypeScript knows exactly which shape you are holding.

```
type Shape =  
  | { kind: 'circle'; radius: number }  
  | { kind: 'square'; side: number }  
  | { kind: 'rect'; w: number; h: number };
```

kind is the DISCRIMINANT: a literal type unique to each member.



Everything that narrows

typeof x === 'string'
for primitives
x instanceof Date
for class instances
'radius' in s
for plain object shapes
Array.isArray(x)
for arrays
if (x) / if (x === null)
truthiness and null
x.kind === 'circle'
the discriminant: the
one that scales